

NovaTech Q Exploration Log

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Q Exploration Questions

For each question, record Q's answer, then cross-check it against the corresponding dashboard visual. Use the Match and Notes columns to document whether the answer is supported by the dashboard.

#	Question Asked	Q's Answer (Summarize)	Dashboard Visual Used to Cross-Check	Dashboard Shows	Match?	Notes
1	Which campaign channel has the highest conversion rate?	Q ranked Partner Referral highest, but reported nonsensical summed values (3,603.78, total 11,591.60) instead of a true 0-1 ratio	Marketing Funnel — Leads by Channel / Campaign Performance table	Partner Referral has the most leads (757) and reasonable campaign_roi; no direct "conversion rate" metric on the dashboard itself	Yes Partial	Q's <i>ranking</i> (Partner Referral highest) is directionally plausible and matches the lead-volume leader on the dashboard, but the displayed Conversion Rate values are inflated by Q summing a per-row ratio across duplicate joined rows rather than computing one true rate the same row-multiplication issue handled manually in the dashboard build. The underlying per-deal value (visible in the raw results table, e.g., 0.18) appears correct; only the chart/KPI aggregation is wrong.
2	What is the average deal size by company size?	Q reported overall average deal value = \$1,590.99; grouped by raw Employee Count (85 unique values) rather than a size tier, naming the single account with 9,226 employees (\$9,311) as "highest"	Sales Pipeline — Revenue by Segment and Product / Avg Deal Value by Segment	Company Size Tier shows Small (\$1.67K), Enterprise (\$1.66K), Medium (\$1.51K), Large (\$1.33K) a clean 4-category breakdown	No	Q correctly computed the overall average (\$1,590.99 ≈ matches dashboard's ~\$1.5-1.6K range) but misinterpreted "company size" as the raw Employee Count field instead of the categorical Company Size Tier field, producing an unhelpfully granular, single-account answer instead of a meaningful segment comparison. This suggests Company Size Tier's synonyms may need to include "company size" explicitly to disambiguate from Employee Count.
3	What is the average resolution time for critical vs. low-priority tickets?	Q reported overall average (59.55 hrs) correctly, but grouped by day instead of priority — naming Jun 17, 2024 (99.91 hrs) as highest and Aug 27,	Customer Health — Avg Resolution Time by Priority	Low: ~60 hrs, High: ~60 hrs, Medium: ~57 hrs, Critical: ~55 hrs	No	Q correctly computed the overall average (59.55 matches dashboard's KPI card exactly) but failed to recognize "critical" and "low-priority" as values of the priority field, defaulting to a date-based breakdown instead. This is a comparison/filter-value recognition

		2023 (13 hrs) as lowest				gap rather than a field-name ambiguity Q didn't connect the phrase "critical vs. low-priority" to the Priority dimension's actual values.
4	What are the top 10 accounts by support ticket volume, and what is their total deal revenue?	Top 10 accounts = 1,513 unique tickets (plausible) and \$39,550,986 total deal value (implausible)	Customer Health — Support Ticket Volume by Account chart, and At-Risk Accounts scatter (deal_value average ranges ~\$500-\$10K per account)	Individual account deal values in the risk scatter top out around \$9.5K (ACCT-045); no account plausibly totals into the millions	No	Ticket count uses Count Distinct correctly, but deal revenue appears to be a raw SUM of deal_value across every duplicated row from the CRM→Marketing→Support join the same row-multiplication error the dashboard build specifically corrected for using Count Distinct and AVERAGE/sumOver(..., PRE_AGG). This is the clearest, highest-stakes accuracy gap found in Q exploration: any revenue total Q reports on the unified dataset should be treated with suspicion unless independently verified.
5	Are there any campaigns where we spent more than we earned back?	Yes — all 5 (of 6) unique campaign channels shown have negative total Campaign ROI; Direct Mail is "highest" at -1,592.32; overall total -59,612.96	Marketing Funnel — Campaign Performance table (campaign_roi column)	All 6 campaigns show negative ROI, ranging roughly -0.9 to -1.0	Yes Partial	The directional answer is correct yes, every campaign is currently unprofitable and the identity of the "best" (least-negative) channel is plausible. But the magnitude (-1,592.32 vs. the dashboard's true ~-0.9 to -1.0 range) is inflated by summing campaign_roi across duplicated join rows rather than averaging or using a distinct per-campaign value the same summation bug seen in Rows 1 and 4. Also notable: Q surfaced only 5 channels, not 6 worth double-checking whether one was silently dropped by a filter or grouping issue.

Reflection

• Where did Q agree with the dashboard?

- Q's overall averages/aggregate baselines were consistently accurate (59.55 hrs resolution time, \$1,590.99 avg deal value) single, non-grouped totals were reliable.
- Directional/ranking answers were often correct even when magnitudes were wrong (Partner Referral highest conversion, Direct Mail least-negative ROI, Mobile App-tier product areas having similar resolution times).

- **Where did Q disagree or struggle? Why?**

- **Field ambiguity:** Q sometimes picked a technically valid but business-wrong field when two fields had overlapping synonyms (Contact Channel vs. Product Area; Employee Count vs. Company Size Tier).
- **Value recognition gap:** Q didn't connect plain-language category values ("critical," "low-priority") to the actual priority field without being asked more literally.
- **Row-multiplication/aggregation bug:** the most serious and recurring issue any question requiring a **sum** on the unified dataset (total revenue, total ROI, total conversion rate) was inflated by orders of magnitude, because Q summed values across duplicated joined rows instead of computing one true value per entity. This happened in 3 of 5 questions (Rows 1, 4, 5).

- **When would you use Q vs. the dashboard to answer a business question?**

- Use Q for quick directional questions, rankings, and simple filtered lookups on a single dataset it's fast and generally reliable there.
- Use the dashboard (not Q) for any question involving **totals, sums, or precise dollar figures** on the unified dataset these require the careful Count Distinct/PRE_AGG handling that the dashboard's calculated fields already apply correctly, but that Q's on-the-fly aggregation does not consistently replicate.

NovaTech Revenue Intelligence Topic

PINBOARD

SHARE

FEEDBACK

Which campaign channel has the highest conversion rate?

ASK

Mark as verified

Interpreted as: Total Conversion Rate by Campaign Channel.

The total Conversion Rate across all campaign channels is 11,591.60, spanning 5 unique campaign channels. Among these, Partner Referral leads as the top campaign platform with a Conversion Rate of 3,603.78. As of January 25, 2025, the year-to-date total Conversion Rate experienced a significant decline of 95.33%, dropping by 1,066.13 from 1,118.4 to 52.27. Similarly, the month-to-date total Conversion Rate as of January 25, 2025 decreased by 71.63%, falling by 131.96 from 184.24 to 52.27.

REVIEW FOR ACCURACY

Total Conversion Rate by Campaign Channel

Partner Referral

3.6K

Organic Search

2.93K

Email

2.5K

Paid Social

2.11K

Direct Mail

0.44K

Total Conversion Rate

11,591.6

Unique number of Campaign Channel

5

Total Conversion Rate by Campaign Channel and day

SHOWING TOP 200 IN DEAL_CREATED_DATE and TOP 5 IN CAMPAIGN_CHANNEL

Funnel Stage, Product Area, Sales Rep, Sales Manager, Deal Created Date, Conversion Rate and Campaign Channel

Product Area	Funnel Stage	Campaign Channel	Deal Created Date	Sales Manager	Sales Rep	Conversion Rate
Analytics Dashboard	Closed Won	Partner Referral	Apr 17, 2024	Michael Chen	Amira Diallo	0.18
Analytics Dashboard	Closed Won	Partner Referral	May 13, 2024	Michael Chen	Amira Diallo	0.18
Analytics Dashboard	Closed Won	Paid Social	May 17, 2024	Michael Chen	Amira Diallo	0.18

View: 500 items

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NovaTech Revenue Intelligence Topic

PINBOARD

SHARE

FEEDBACK

What is the average deal size by company size?

ASK

Mark as verified

Interpreted as: Average of Deal Value by Employee Count.

The overall average Deal Value across all records is 1,590.99, spanning 85 unique Employee Count values. Among these, the Employee Count of 9,226 has the highest average deal amount at 9,311.

REVIEW FOR ACCURACY

Average of Deal Value

1,590.99

Unique number of Employee Count

85

Average of Deal Value by Employee Count

9,226

9.31K

299

4.52K

2,822

3.04K

6,290

2.89K

5,113

2.85K

34,288

2.71K

3,492

2.6K

3,502

2.48K

1,016

2.46K

853

2.44K

4,540

2.39K

Sales Rep, Sales Manager, Funnel Stage, Campaign Channel, Product Area, Deal Created Date, Employee Count and Deal Value

Product Area	Funnel Stage	Campaign Channel	Deal Value	Deal Created Date	Sales Manager	Sales Rep	Employee Count
Analytics Dashboard	Lead	Organic Search	27,385	Feb 21, 2024	Amanda Foster	Grace Adeyemi	9,226
Authentication	Lead	Organic Search	27,385	Feb 21, 2024	Amanda Foster	Grace Adeyemi	299
Billing	Lead	Organic Search	27,385	Feb 21, 2024	Amanda Foster	Grace Adeyemi	2,822

View: 500 items

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NovaTech Q Exploration Log

What is the average resolution time for critical vs. low-priority tickets?

👤

ASK

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Mark as verified

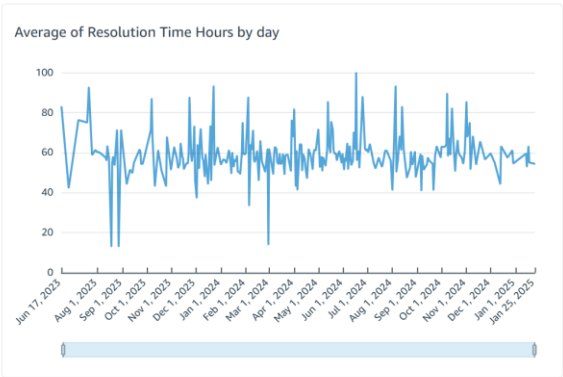
Interpreted as: Average of Resolution Time Hours.

✎

See Why

The average Resolution Time Hours is 59.55. Among all days, Jun 17, 2024 recorded the highest average Resolution Time Hours at 99.91, while Aug 27, 2023 had the lowest at 13.

REVIEW FOR ACCURACY ⓘ



Product Area, Contact Channel, Sales Rep, Sales Manager, Campaign Channel, Deal Created Date and Resolution Time Hours

Resolution Time H...	Contact Channel	Product Area	Campaign Channel	Deal Created Date	Sales Manager	Sales Rep
163	Email	Data Pipeline	Direct Mail	May 24, 2024	Rachel Torres	Carlos Rivera
163	Email	Data Pipeline	Direct Mail	Sep 20, 2024	Rachel Torres	Carlos Rivera
163	Email	Data Pipeline	Direct Mail	Nov 29, 2023	Michael Chen	Finn O'Brien

View: 500 items << < 1 of 77 > >>

What are the top 10 accounts by support ticket volume, and what is their total deal revenue?

👤

ASK

↩️ ↪️

Mark as verified

Interpreted as: Top 10 Account Id by unique number of Ticket Id and total Deal Value.

✎

Across a total of 85 unique accounts, the top 10 accounts by unique number of Ticket Id collectively account for 1,513 unique tickets and a total Deal Value of 39,550,986.

REVIEW FOR ACCURACY ⓘ

Did you mean...

Top 10 Account Ids by Ticket Id and total Revenue Attributed.

Account Id by unique number of Ticket Id and total Deal Value TOP 10

Account Id	Deal Value	Ticket Id
ACCT-041	13,601,148	334
ACCT-035	3,014,460	180
ACCT-076	8,593,200	176
ACCT-043	2,543,454	171
ACCT-060	4,481,520	142
ACCT-036	2,226,000	140
ACCT-020	654,654	109
ACCT-072	2,482,020	90
ACCT-025	1,209,688	89
ACCT-018	744,842	83

Unique number of Tic... TOP 10 ACCOUNT IDS BY ... 1,513	Total Deal Value TOP 10 ACCOUNT IDS BY ... 39,550,986	Unique number of Acc... 85
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Unique number of Account Id

Total Deal Value by Account Id and Ticket Id TOP 10 ACCOUNT IDS BY TICKET ID

Ticket Id	Account Id	Deal Value
TKT-100258	ACCT-018	8,974
TKT-100283	ACCT-076	48,825
TKT-101055	ACCT-041	40,722
TKT-101657	ACCT-060	31,560
TKT-105621	ACCT-060	31,560

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Product Area, Resolution Time Hours, Contact Channel, Employee Count, Sales Rep, Sales Manager, Funnel Stage, Campaign Channel, Account Id, Ticket Id ... TOP 10 ACCOUNT IDS BY TICKET ID

Resolution Time Hours	Contact Channel	Product Area	Ticket Id	Account Id	Funnel Stage	Campaign Channel	Deal Value
100	Email	Analytics Dashboard	TKT-999474	ACCT-036	Lead	Organic Search	0
100	Email	Analytics Dashboard	TKT-999474	ACCT-036	Prospect	Email	0

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Are there any campaigns where we spent more than we earned back?

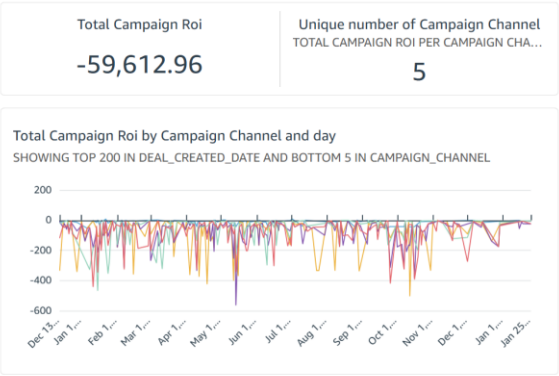
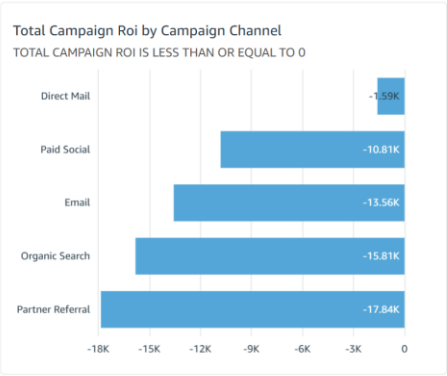
ASK

Mark as verified

Interpreted as: **Total Campaign Roi by Campaign Channel where total Campaign Roi is less than or equal to 0.**

Among campaign channels where total Campaign Roi is less than or equal to 0, the overall total Campaign Roi stands at -59,612.96, spanning 5 unique campaign channels. Of these, Direct Mail recorded the highest total Campaign Roi at -1,592.32. Additionally, the total campaign cost across these channels is -47,866.57. On a year-to-date basis as of Jan 25, 2025, total campaign cost increased by 95.29% (5,437.01), moving from -5,705.68 to -268.67. Similarly, on a month-to-date basis as of Jan 25, 2025, total campaign cost increased by 71.18% (663.59), rising from -932.26 to -268.67.

REVIEW FOR ACCURACY



Funnel Stage, Sales Rep, Sales Manager, Product Area, Deal Created Date, Campaign Roi and Campaign Channel

Product Area	Campaign Roi	Funnel Stage	Campaign Channel	Deal Created Date	Sales Manager	Sales Rep
Mobile App	6.05	Closed Won	Organic Search	Oct 7, 2023	Sarah Mitchell	Derek Johnson
Mobile App	6.05	Closed Won	Organic Search	Feb 4, 2024	David Kim	Marcus Webb
Mobile App	6.05	Closed Won	Organic Search	Jul 21, 2023	David Kim	Owen McCarthy

View: 500 items

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